Swindon Eastern Villages

Technical Note : White Hart Interchange

22nd June 2015

1 Introduction

Brookbanks Consulting Limited (BCL) is appointed by the Swindon Eastern Village Consortium (SEVC) to complete a Transportation Assessment (TA) and a Travel Plan (TP) for a proposed mixed use development on land to the east of Swindon.

The purpose of the Technical Note is to identify the mitigation strategy for the White Hart Interchange.

2 Junction Assessment Review – White Hart

Background

As part of evidence prepared in support of the Swindon Eastern Villages development proposals, BCL has worked in conjunction with Swindon Borough Council (SBC) and Highways England (HE) representatives to work toward agreement on all highways matters, including trip generation forecasts, traffic modelling methodology and results, together with the production of a package of mitigation measures to enhance the highway network and accessibility of the area while mitigating for transport impacts of the development proposals.

The potential impacts on the highway network have been tested through the Swindon area wide SATURN traffic model together with the A419 / A420 PARAMICS traffic model. This has resulted in the identification of a range of offsite interventions that will delivered by the development, including at White Hart.

The TA identifies that to mitigate the impact of the current application, several approach arms to White Hart will require signalisation.

The current Local Plan indicates that the Swindon Borough Council preferred solution includes the provision of northbound on and off slips and amending the layout to reflect a 'diamond interchange'.

Modelling Protocol

The scope of the TA was discussed with SBC and HE. These discussions informed the modelling protocol that used traffic flows generated by the Swindon wide Saturn traffic model and inputted into the Paramics A420 / A419 traffic model. The models were established for 2016 and 2026.

The EV development will be delivered over a number of years. The Paramics model assessed the impact of the EV in three main phases, as indicated below:

- Phase 1 – 2,500 north of the A420
- Phase 1A – as Phase 1 plus employment area south of A420
- Phase 2 – as Phase 1A plus the remaining EV south of A420, compliant to Local Plan

For each phase tested, the models included a range of interventions. For Phase 1, a range of interventions were included that are required to mitigate the impact of the development. For Phase 2, the interventions identified within the Local Plan were assumed.

In relation to the White Hart Interchange, for Phase 1 the mitigation included the signalisation of the majority of the approach arms. For Phase 2, the diamond interchange was included.

Alternative White Hart Interchange Intervention

Through discussions with SBC, BCL have reviewed the White Hart Interchange to determine the extent of improvements that are required at the junction to mitigate the impact of Phase 2, whilst maintaining the current form of the junction.

Prior to any formal testing in the Paramics model, a Linsig model was set up to test any potential intervention identified. The use of Linsig is a logical first step, as this will identify if the identified alternative intervention is likely to work, prior to the process of altering the more formal Paramics model.

Turning flows for the diamond interchange were provided following the Phase 2 tests. This indicated the total flows on the approach arms. As these were provided for the diamond interchange layout, these flows were manipulated to provide equivalent flows needed for the existing layout. This was achieved in the following way:

Step One: The turning proportions and total flow on each arm was calculated based on the 2026 Phase 1 were identified.

Step Two: The 2026 Phase 2 total flows on each arm were established from the results of the Phase two modelling results.

Step Three: The resultant 2026 Phase Two traffic flows were assigned to the existing layout based on the turning proportions identified in Step One.

Step Four: The resultant 2026 Phase Two traffic flows were compared against the 2026 Phase One turning proportions as a sense check.

Following the identification of the traffic flows, the design of the necessary intervention at White Hart was considered. An improvement scheme was developed that included the increasing the width of the circulatory carriageway together with widening of some of the approach arms. The proposed scheme is contained in the Appendix.

Assessment Results

The results of the junction assessment are presented below. There are several approach arms that have more than one lane, in this case the lane with the highest Degree of Saturation is reported, with the full output contained in the Appendix.

Link	Morning Peak		Evening Peak	
	Degree of Saturation	Queue Length (m)	Degree of Saturation	Queue Length (m)
Eastern A420 arm	84.7%	68	72.9%	30
Merlin Way approach	75.9%	34	79.1%	37
Oxford Road approach	76.3%	34	80.9%	45
A419 southbound offslip	42.7%	16	88.0%	58
Ermin Way	81.9%	51	85.3%	52

Figure 2a: White Hart Interchange, 2026 Full Signalisation

This demonstrates that the junction will operate within acceptable limits with the highest degree of saturation predicted to be 88.0%, with 90% typically assumed to reflect an acceptable capacity threshold.

Linsig also predicts queue lengths across the junction. This demonstrates that the queuing will not block back to affect the operation of any adjacent junctions.

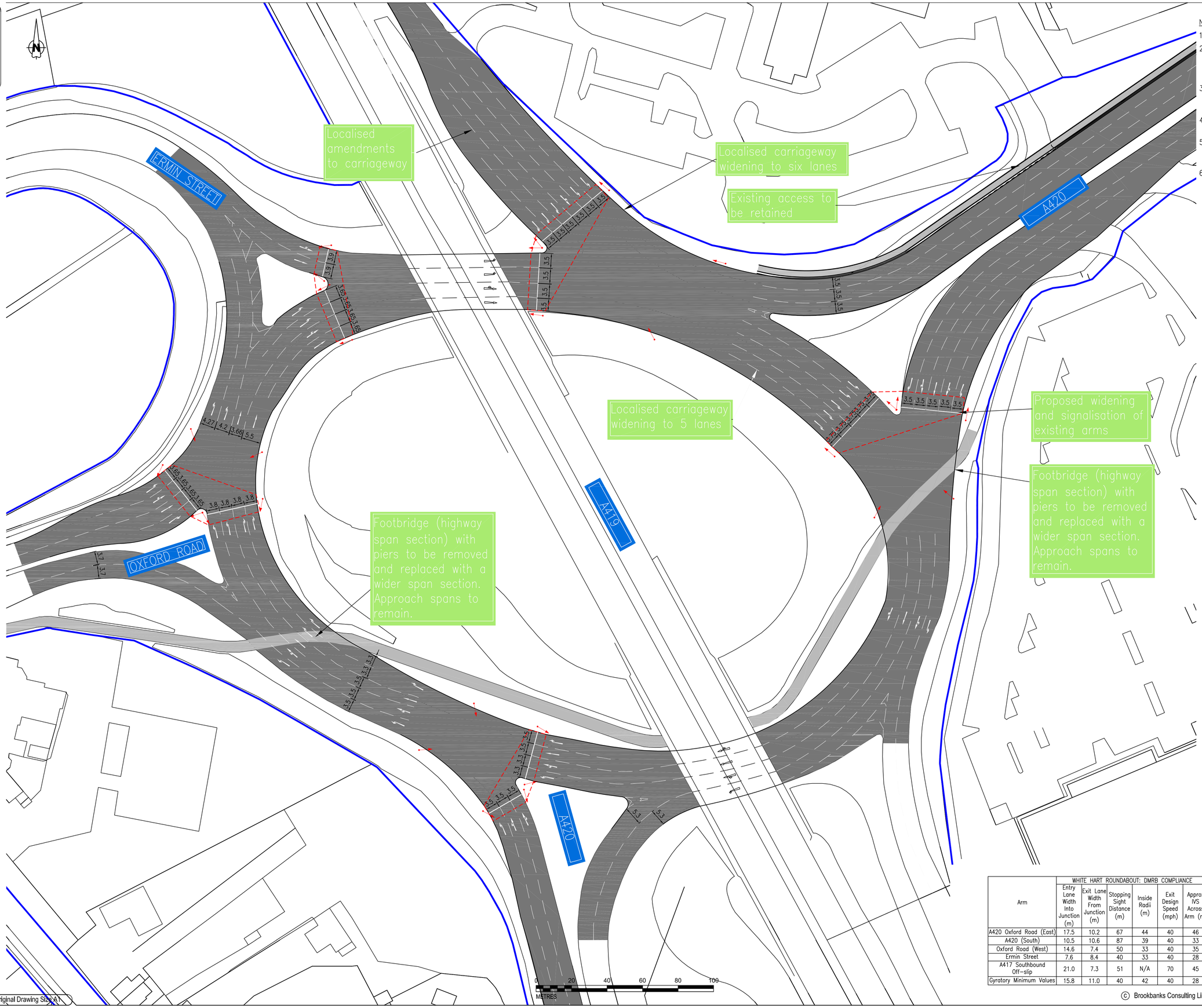
The figure below provides a comparison between the Paramics 2026 reference case, as reported in Figure 11aa of the TA, and the Linsig assessment reported above.

Link	Linsig queuing results		Paramics queuing results	
	Morning Peak	Morning Peak	Morning Peak	Evening Peak
Eastern A420 arm	68	30	268	166
Merlin Way approach	34	37	315	347
Oxford Road approach	34	45	49	306
A419 southbound offslip	16	58	556	137
Ermin Way	51	52	201	29

Figure 2b: Comparison of results

The above comparison identifies that the conversion to full signalisation together with road widening will significantly improve the operation of the junction. This highlights that the proposed scheme will deliver a realistic alternative to the diamond interchange.

1335/HL/54



Notes:

1. Do not scale from this drawing.
2. This drawing has been based on survey data provided by a third party. Brookbanks Consulting Ltd cannot be held responsible for the accuracy of this information.
3. The roundabout and links have been designed in accordance with the relevant DMRB standards TD 9/93 and TD 16/07.
4. The markings have been designed in accordance with the Traffic Signs Manual Chapter 5.
5. The road signs have been designed and positioned in accordance with the Traffic Signs Manual Chapters 3 and 7. To be placed at 400mm min. from kerb.
6. The signalised roundabout has been designed in accordance with TD 50/04 The Geometric Layout of Signal Controlled Junctions and Signalised Roundabouts. The table below represents the design's compliance with the standard.

Arm	WHITE HART ROUNDABOUT: DMRB COMPLIANCE					
	Entry Lane Width Into Junction (m)	Exit Lane Width From Junction (m)	Stopping Sight Distance (m)	Inside Radii (m)	Exit Design Speed (mph)	Approx IVS Across Arm (m)
A420 Oxford Road (East)	17.5	10.2	67	44	40	46
A420 (South)	10.5	10.6	87	39	40	33
Oxford Road (West)	14.6	7.4	50	33	40	35
Ermin Street	7.6	8.4	40	33	40	28
A417 Southbound Off-slip	21.0	7.3	51	N/A	70	45
Gyratory Minimum Values	15.8	11.0	40	42	40	28

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Rev.	Revision Details	Drawn	Checked	Approved	Date
	FOR PRELIMINARY REVIEW	PAB			11.06.15

Issue Status			Approved	Date
Drawn	MDM	Checked	AE	Date JUN 15

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Swindon EDAG

Swindon EDA

WHITE HART SIGNALISATION PROPOSED LAYOUT

Scale at A1	Drawing No.	Rev.
1:500	1335/HL/54	-

Original Drawing Size: A1

Linsig Report

User and Project Details

Project:	**Swindon EDA**
Title:	**Signalled Gyratory**
Location:	Swindon
File name:	1335L01-A449-A420 White Hart Gyratory 1335HL54 June 2015.lsg3x
Author:	SMT
Company:	Brookbanks Consulting Ltd

Linsig Report
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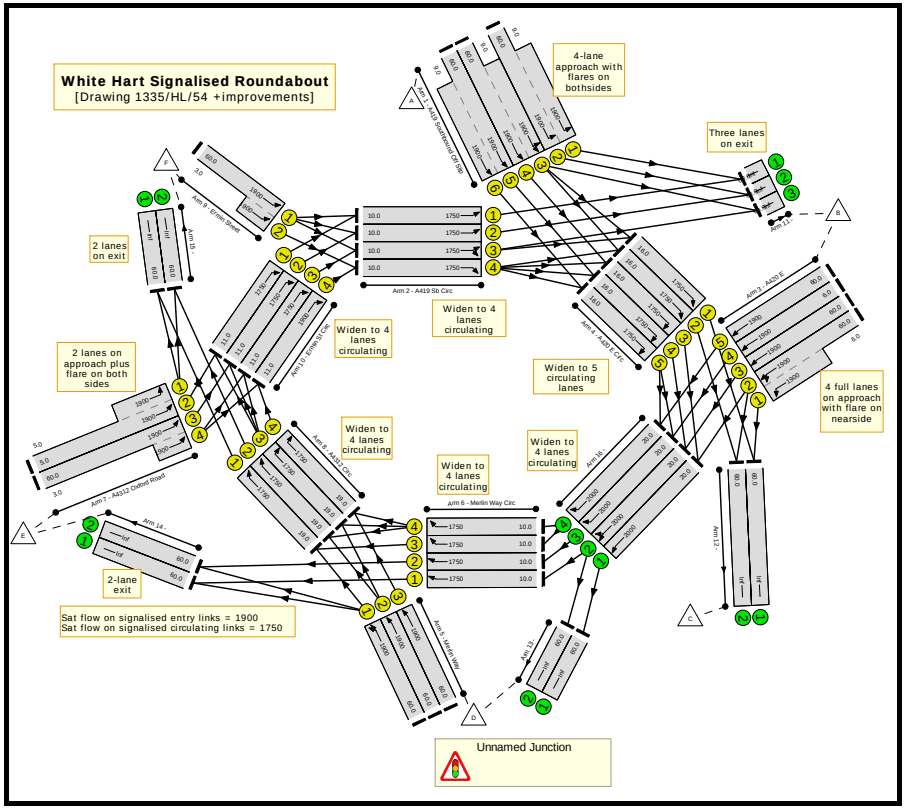
Junction Layout Diagram



Linsig Report
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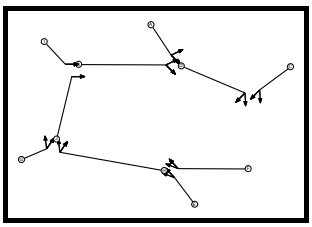
Phase Diagram



Phase Intergreens Matrix

		Starting Phase									
		A	B	C	D	E	F	G	H	I	J
Terminating Phase	A		5	-	-	-	-	-	-	-	-
	B	6		-	-	-	-	-	-	-	-
	C	-	-		5	-	-	-	-	-	-
	D	-	-	6		-	-	-	-	-	-
	E	-	-	-	-		5	-	-	-	-
	F	-	-	-	-	6		-	-	-	-
	G	-	-	-	-	-	-		5	-	-
	H	-	-	-	-	-	-	6		-	-
	I	-	-	-	-	-	-	-	-		5
	J	-	-	-	-	-	-	-	-	6	

Lane Input Data

Junction: Unnamed Junction

Lane	Lane Type	Phases	Start Disp.	End Disp.	Physical Length (PCU)	Sat Flow Type	Def User Saturation Flow (PCU/Hr)	Lane Width (m)	Gradient	Nearside Lane	Turns	Turning Radius (m)
1/1 (A419 Southbound Off Slip)	U	A	2	3	9.0	User	1900	-	-	-	-	-
1/2 (A419 Southbound Off Slip)	U	A	2	3	60.0	User	1900	-	-	-	-	-
1/3 (A419 Southbound Off Slip)	U	A	2	3	9.0	User	1900	-	-	-	-	-
1/4 (A419 Southbound Off Slip)	U	A	2	3	60.0	User	1900	-	-	-	-	-
1/5 (A419 Southbound Off Slip)	U	A	2	3	60.0	User	1900	-	-	-	-	-
1/6 (A419 Southbound Off Slip)	U	A	2	3	9.0	User	1900	-	-	-	-	-
2/1 (A419 Sb Circ)	U	B	2	3	10.0	User	1750	-	-	-	-	-
2/2 (A419 Sb Circ)	U	B	2	3	10.0	User	1750	-	-	-	-	-
2/3 (A419 Sb Circ)	U	B	2	3	10.0	User	1750	-	-	-	-	-
2/4 (A419 Sb Circ)	U	B	2	3	10.0	User	1750	-	-	-	-	-
3/1 (A420 E)	U	C	2	3	6.0	User	1900	-	-	-	-	-
3/2 (A420 E)	U	C	2	3	60.0	User	1900	-	-	-	-	-
3/3 (A420 E)	U	C	2	3	60.0	User	1900	-	-	-	-	-
3/4 (A420 E)	U	C	2	3	6.0	User	1900	-	-	-	-	-
3/5 (A420 E)	U	C	2	3	60.0	User	1900	-	-	-	-	-
4/1 (A420 E Circ)	U	D	2	3	16.0	User	1750	-	-	-	-	-
4/2 (A420 E Circ)	U	D	2	3	16.0	User	1750	-	-	-	-	-
4/3 (A420 E Circ)	U	D	2	3	16.0	User	1750	-	-	-	-	-
4/4	U	D	2	3	16.0	User	1750					

Linsig Report
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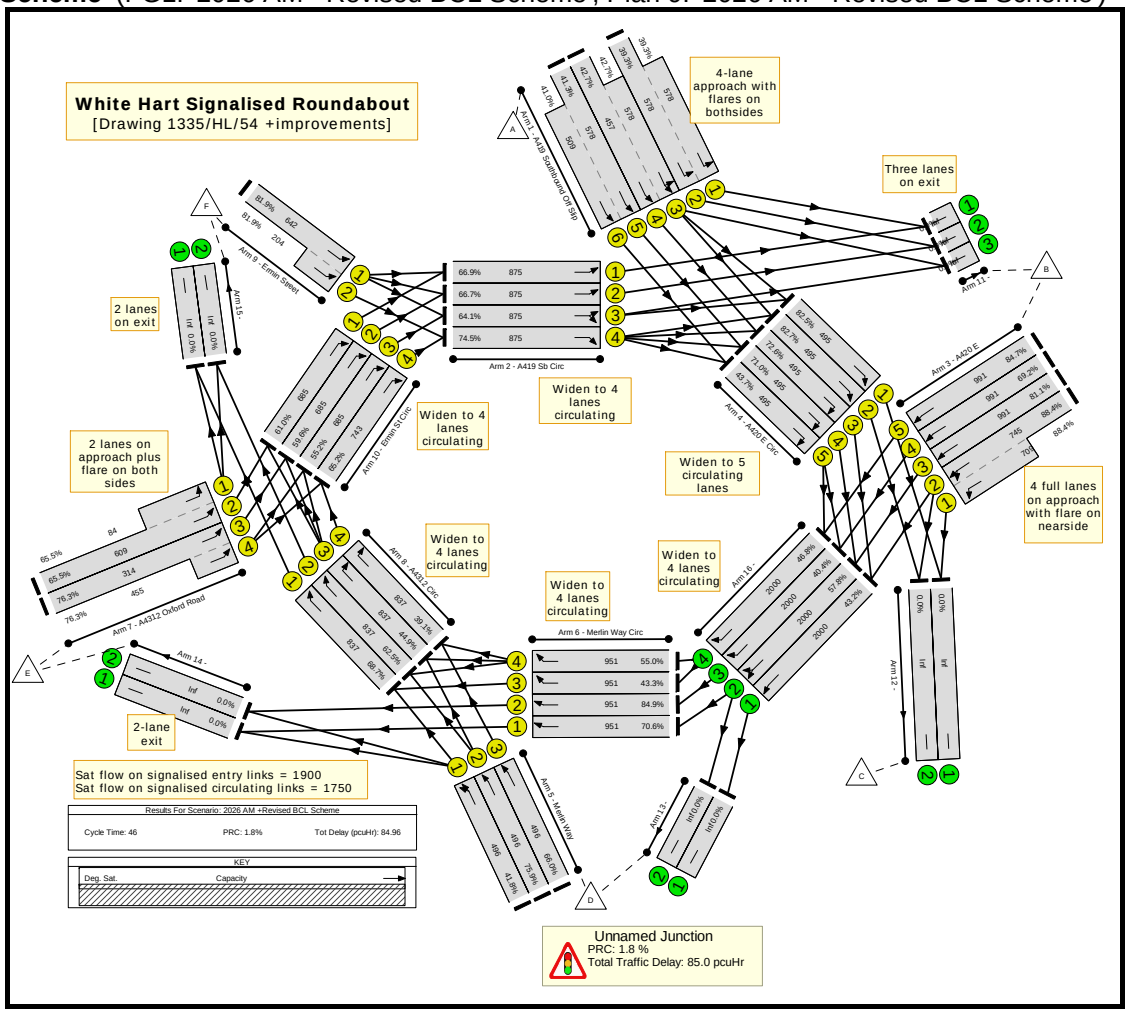
Scenario 1: '2026 AM +Revised BCL Scheme' (FG1: '2026 AM +Revised BCL Scheme', Plan 9: '2026 AM +Revised BCL Scheme')



Linsig Report
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Link Results

Item	Lane Description	Lane Type	Full Phase	Total Green (s)	Demand Flow (pcu)	Sat Flow (pcu/Hr)	Capacity (pcu)	Deg Sat (%)	Av. Delay Per PCU (s/pcu)	Mean Max Queue (pcu)	Average Excess Queue (pcu)
Network: Signalled Gyratory	-	-	-	-	-	-	-	**88.4%**	-	-	-
Unnamed Junction	-	-	-	-	-	-	-	**88.4%**	-	-	-
1/2+1/1	A419 Southbound Off Slip Left	U	A	13	454	1900:1900	578+578	39.3 : 39.3%	15.2	2.6	0.00
1/4+1/3	A419 Southbound Off Slip Ahead Left	U	A	13	442	1900:1900	457+578	42.7 : 42.7%	15.7	2.8	0.00
1/5+1/6	A419 Southbound Off Slip Ahead	U	A	13	448	1900:1900	578+509	41.3 : 41.0%	15.4	2.7	0.00
2/1	A419 Sb Circ Ahead	U	B	22	585	1750	875	66.9%	11.0	4.0	0.00
2/2	A419 Sb Circ Ahead	U	B	22	584	1750	875	66.7%	10.6	4.4	0.00
2/3	A419 Sb Circ Ahead Ahead2	U	B	22	561	1750	875	64.1%	10.5	4.4	0.00
2/4	A419 Sb Circ Ahead	U	B	22	652	1750	875	74.5%	11.9	4.5	0.00
3/2+3/1	A420 E Left Ahead	U	C	23	1285	1900:1900	745+709	88.4 : 88.4%	18.2	10.0	0.00
3/3	A420 E Ahead	U	C	23	804	1900	991	81.1%	18.5	10.6	0.00
3/4	A420 E Ahead	U	C	23	686	1900	991	69.2%	14.1	7.6	0.00
3/5	A420 E Ahead	U	C	23	840	1900	991	84.7%	20.9	11.8	0.00
4/1	A420 E Circ Ahead	U	D	12	408	1750	495	82.5%	31.2	6.7	0.14
4/2	A420 E Circ Ahead	U	D	12	409	1750	495	82.7%	33.3	5.0	0.00
4/3	A420 E Circ Right	U	D	12	359	1750	495	72.6%	28.6	5.7	0.00
4/4	A420 E Circ Right	U	D	12	351	1750	495	71.0%	27.4	5.5	0.00
4/5	A420 E Circ Right	U	D	12	216	1750	495	43.7%	20.2	3.1	0.00
5/1	Merlin Way Ahead Ahead2	U	E	11	207	1900	496	41.8%	20.3	2.5	0.00
5/2	Merlin Way Ahead	U	E	11	376	1900	496	75.9%	30.3	5.9	0.00
5/3	Merlin Way Ahead	U	E	11	327	1900	496	66.0%	25.7	4.7	0.00

6/1	Merlin Way Circ Ahead	U	F	24	671	1750	951	70.6%	11.2	4.4	0.00
6/2	Merlin Way Circ Ahead	U	F	24	807	1750	951	84.9%	16.4	6.1	0.00
6/3	Merlin Way Circ Right	U	F	24	412	1750	951	43.3%	7.4	1.7	0.00
6/4	Merlin Way Circ Right	U	F	24	523	1750	951	55.0%	8.7	2.6	0.00
7/2+7/1	A4312 Oxford Road Ahead Left	U	G	14	454	1900:1900	609+84	65.5 : 65.5%	20.4	5.3	0.00
7/3+7/4	A4312 Oxford Road Ahead	U	G	14	587	1900:1900	314+455	76.3 : 76.3%	22.4	5.9	0.00
8/1	A4312 Circ Ahead	U	H	21	575	1750	837	68.7%	14.1	6.6	0.00
8/2	A4312 Circ Ahead	U	H	21	523	1750	837	62.5%	18.0	7.5	0.00
8/3	A4312 Circ Right	U	H	21	376	1750	837	44.9%	10.2	1.8	0.00
8/4	A4312 Circ Right	U	H	21	327	1750	837	39.1%	8.7	1.3	0.00
9/1+9/2	Ermin Street Ahead	U	I	18	693	1900:1900	642+204	81.9 : 81.9%	22.6	8.9	0.00
10/1	Ermin St Circ Right	U	J	17	418	1750	685	61.0%	10.4	1.7	0.00
10/2	Ermin St Circ Right	U	J	17	408	1750	685	59.6%	12.3	3.3	0.00
10/3	Ermin St Circ Right	U	J	17	378	1750	685	55.2%	12.4	3.3	0.00
10/4	Ermin St Circ Right	U	J	17	485	1900	743	65.2%	15.3	5.9	0.00
16/1	Ahead	U	-	-	863	2000	2000	43.2%	1.6	0.4	0.00
16/2	Right Ahead	U	-	-	1155	2000	2000	57.8%	2.1	0.7	0.00
16/3	Right	U	-	-	807	2000	2000	40.4%	1.5	0.3	0.00
16/4	Right	U	-	-	935	2000	2000	46.8%	1.7	0.4	0.00

C1	Stream: 1	PRC for Signalled Lanes (%):	20.8	Total Delay for Signalled Lanes (pcuHr):	13.07	Cycle Time (s):	46
C1	Stream: 2	PRC for Signalled Lanes (%):	1.8	Total Delay for Signalled Lanes (pcuHr):	32.24	Cycle Time (s):	46
C1	Stream: 3	PRC for Signalled Lanes (%):	6.1	Total Delay for Signalled Lanes (pcuHr):	14.54	Cycle Time (s):	46
C1	Stream: 4	PRC for Signalled Lanes (%):	17.9	Total Delay for Signalled Lanes (pcuHr):	12.96	Cycle Time (s):	46
C1	Stream: 5	PRC for Signalled Lanes (%):	9.9	Total Delay for Signalled Lanes (pcuHr):	10.31	Cycle Time (s):	46
		PRC Over All Lanes (%):	1.8	Total Delay Over All Lanes(pcuHr):	84.96		

Linsig Report
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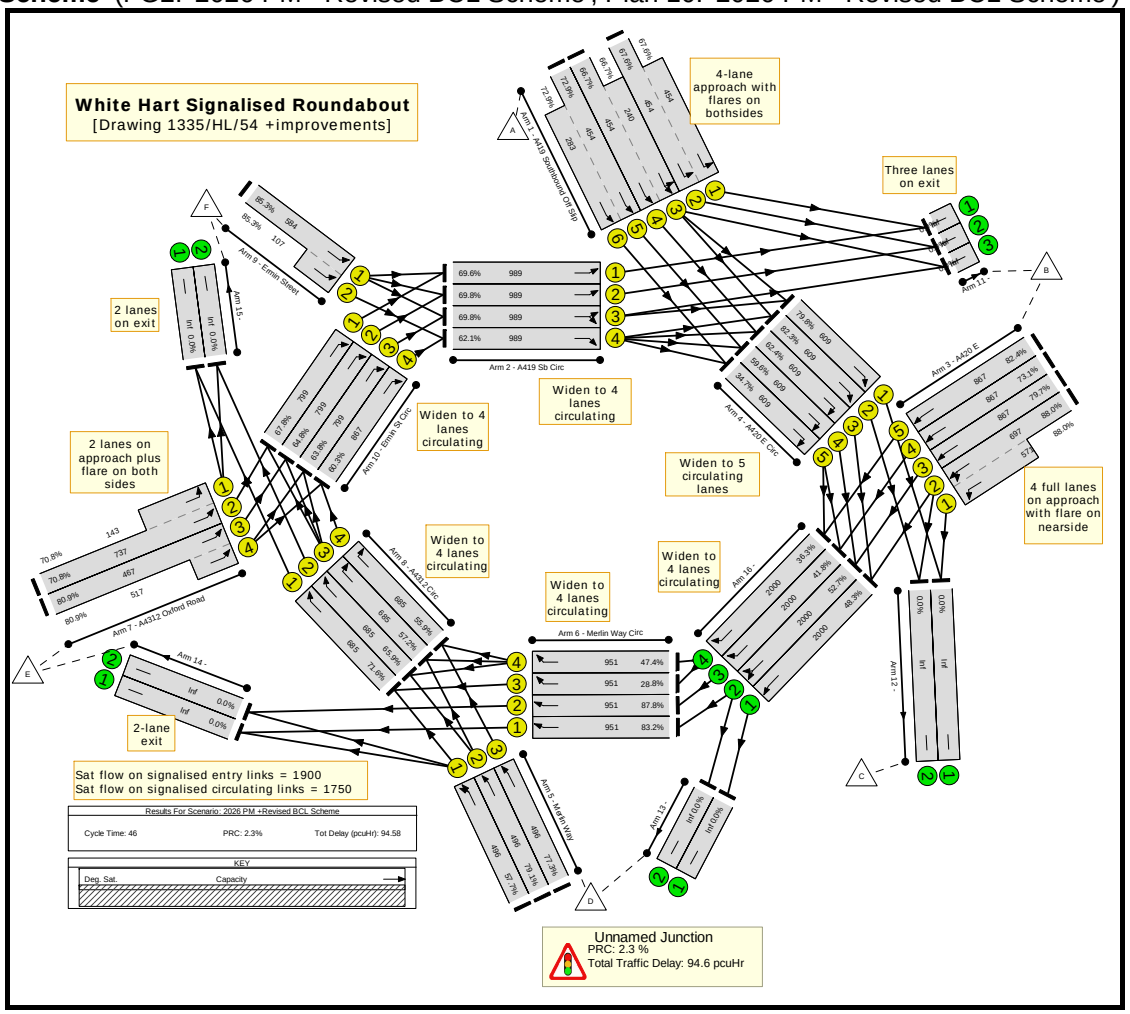
Scenario 2: '2026 PM +Revised BCL Scheme' (FG2: '2026 PM +Revised BCL Scheme', Plan 10: '2026 PM +Revised BCL Scheme')



Linsig Report
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Link Results

Item	Lane Description	Lane Type	Full Phase	Total Green (s)	Demand Flow (pcu)	Sat Flow (pcu/Hr)	Capacity (pcu)	Deg Sat (%)	Av. Delay Per PCU (s/pcu)	Mean Max Queue (pcu)	Average Excess Queue (pcu)
Network: Signalled Gyratory	-	-	-	-	-	-	-	**88.0%**	-	-	-
Unnamed Junction	-	-	-	-	-	-	-	**88.0%**	-	-	-
1/2+1/1	A419 Southbound Off Slip Left	U	A	10	614	1900:1900	454+454	67.6 : 67.6%	22.0	4.5	0.00
1/4+1/3	A419 Southbound Off Slip Ahead Left	U	A	10	463	1900:1900	454+240	66.7 : 66.7%	23.1	4.4	0.00
1/5+1/6	A419 Southbound Off Slip Ahead	U	A	10	537	1900:1900	454+283	72.9 : 72.9%	24.6	5.2	0.00
2/1	A419 Sb Circ Ahead	U	B	25	688	1750	989	69.6%	8.8	3.6	0.00
2/2	A419 Sb Circ Ahead	U	B	25	690	1750	989	69.8%	9.1	4.5	0.00
2/3	A419 Sb Circ Ahead Ahead2	U	B	25	690	1750	989	69.8%	9.1	4.6	0.00
2/4	A419 Sb Circ Ahead	U	B	25	614	1750	989	62.1%	6.6	2.8	0.00
3/2+3/1	A420 E Left Ahead	U	C	20	1115	1900:1900	697+571	88.0 : 88.0%	21.0	10.1	0.00
3/3	A420 E Ahead	U	C	20	691	1900	867	79.7%	20.7	9.4	0.00
3/4	A420 E Ahead	U	C	20	634	1900	867	73.1%	17.8	7.9	0.00
3/5	A420 E Ahead	U	C	20	715	1900	867	82.4%	22.4	10.2	0.00
4/1	A420 E Circ Ahead	U	D	15	486	1750	609	79.8%	21.0	5.4	0.01
4/2	A420 E Circ Ahead	U	D	15	501	1750	609	82.3%	22.3	4.3	0.00
4/3	A420 E Circ Right	U	D	15	380	1750	609	62.4%	22.2	5.5	0.00
4/4	A420 E Circ Right	U	D	15	363	1750	609	59.6%	21.4	5.3	0.00
4/5	A420 E Circ Right	U	D	15	211	1750	609	34.7%	18.0	2.9	0.00
5/1	Merlin Way Ahead Ahead2	U	E	11	286	1900	496	57.7%	23.3	3.9	0.00
5/2	Merlin Way Ahead	U	E	11	392	1900	496	79.1%	32.6	6.4	0.00
5/3	Merlin Way Ahead	U	E	11	383	1900	496	77.3%	31.3	6.1	0.00
6/1	Merlin Way Circ Ahead	U	F	24	791	1750	951	83.2%	17.8	10.7	0.00
6/2	Merlin Way Circ Ahead	U	F	24	835	1750	951	87.8%	21.3	12.7	0.00

6/3	Merlin Way Circ Right	U	F	24	274	1750	951	28.8%	5.3	2.4	0.00
6/4	Merlin Way Circ Right	U	F	24	451	1750	951	47.4%	7.7	5.4	0.00
7/2+7/1	A4312 Oxford Road Ahead Left	U	G	18	623	1900:1900	737+143	70.8 : 70.8%	17.5	6.6	0.00
7/3+7/4	A4312 Oxford Road Ahead	U	G	18	796	1900:1900	467+517	80.9 : 80.9%	19.8	7.8	0.00
8/1	A4312 Circ Ahead	U	H	17	490	1750	685	71.6%	16.4	6.8	0.00
8/2	A4312 Circ Ahead	U	H	17	451	1750	685	65.9%	8.5	2.0	0.00
8/3	A4312 Circ Right	U	H	17	392	1750	685	57.2%	19.0	5.7	0.00
8/4	A4312 Circ Right	U	H	17	383	1750	685	55.9%	18.8	5.5	0.00
9/1+9/2	Ermin Street Ahead	U	I	15	589	1900:1900	584+107	85.3 : 85.3%	30.2	9.1	0.00
10/1	Ermin St Circ Right	U	J	20	542	1750	799	67.8%	9.2	1.9	0.00
10/2	Ermin St Circ Right	U	J	20	518	1750	799	64.8%	12.0	4.5	0.00
10/3	Ermin St Circ Right	U	J	20	510	1750	799	63.8%	15.1	5.8	0.00
10/4	Ermin St Circ Right	U	J	20	523	1900	867	60.3%	19.2	6.7	0.06
16/1	Ahead	U	-	-	966	2000	2000	48.3%	1.7	0.5	0.00
16/2	Right Ahead	U	-	-	1054	2000	2000	52.7%	1.9	0.6	0.00
16/3	Right	U	-	-	835	2000	2000	41.8%	1.5	0.4	0.00
16/4	Right	U	-	-	725	2000	2000	36.3%	1.4	0.3	0.00

C1	Stream: 1	PRC for Signalled Lanes (%):	23.5	Total Delay for Signalled Lanes (pcuHr):	16.68	Cycle Time (s):	46
C1	Stream: 2	PRC for Signalled Lanes (%):	2.3	Total Delay for Signalled Lanes (pcuHr):	29.54	Cycle Time (s):	46
C1	Stream: 3	PRC for Signalled Lanes (%):	2.5	Total Delay for Signalled Lanes (pcuHr):	18.94	Cycle Time (s):	46
C1	Stream: 4	PRC for Signalled Lanes (%):	11.3	Total Delay for Signalled Lanes (pcuHr):	14.76	Cycle Time (s):	46
C1	Stream: 5	PRC for Signalled Lanes (%):	5.5	Total Delay for Signalled Lanes (pcuHr):	12.98	Cycle Time (s):	46
		PRC Over All Lanes (%):	2.3	Total Delay Over All Lanes(pcuHr):	94.58		